Asteroid Risk Prediction and Visualization System

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1. Introduction

Asteroids are celestial objects that orbit the Sun, and some of them pass close to Earth. These near-Earth objects (NEOs) can pose a potential threat if they come too close or collide with our planet. With the advancement of data science and machine learning, it is now possible to analyze asteroid data and predict their risk level.

This project focuses on building a machine learning model that predicts whether an asteroid is **dangerous or safe** based on its distance and velocity. Additionally, a dashboard is created to visualize the data and predictions interactively.

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2. Objectives

The main objectives of this project are:

- To analyze asteroid dataset
- To clean and preprocess real-world data
- To build a machine learning model for prediction
- To classify asteroids as **Safe** or **Dangerous**
- To visualize asteroid data using graphs
- To develop an interactive dashboard for users

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3. Dataset Description

The dataset used in this project contains information about asteroids, including:

- Miss Distance (distance from Earth)
- Relative Velocity (speed of asteroid)
- Other related attributes

This dataset helps in understanding how close and how fast an asteroid is moving, which are key factors in determining its risk.

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4. Methodology

4.1 Data Preprocessing

- Removed missing and irrelevant values
- Identified correct column names dynamically
- Converted data into usable numerical format

4.2 Feature Selection

Two main features were used:

- Distance
- Velocity

These features are most important for determining asteroid risk.

4.3 Risk Classification

A rule-based approach was used initially:

- If distance is very small AND velocity is high → Dangerous
- Otherwise → Safe

4.4 Model Building

A **Decision Tree Classifier** was used:

- Easy to understand
- Works well for classification problems

Steps:

- Split data into training and testing
- Train the model
- Evaluate accuracy

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5. Data Visualization

Different graphs were created:

- Scatter Plot (Distance vs Velocity)
- Histogram (Distance distribution)
- Risk classification visualization

These graphs help in understanding asteroid behavior.

6. Dashboard Development

An interactive dashboard was created using Streamlit.

Features:

- Upload dataset
- Display data
- Show graphs
- Predict asteroid risk
- User-friendly interface

7. Results

- The model successfully classified asteroids
- High accuracy achieved
- Clear visualization of asteroid risk patterns
- Dashboard provides real-time interaction

8. Conclusion

This project demonstrates how machine learning can be used in space-related data analysis. By using simple features like distance and velocity, we can predict asteroid risk effectively.

The dashboard makes it easy for users to interact with the data and understand predictions visually.

9. Future Work

- Use advanced models (Random Forest, Neural Networks)
- Add real-time NASA data
- Improve dashboard UI
- Deploy project online

10. References

- NASA Asteroid Dataset
- Python Libraries: Pandas, Matplotlib, Scikit-learn

- Streamlit Documentation